Ordinary Least Squares

Lecture 2

Data

We start with the multivariate linear regression model,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + u_i$$

which could represent:

- the true data generating process,
- the conditional expectation function,
- an economic model,
- an estimating equation

under different assumptions.

Curcially, the model includes data: for each observation we observation

$$[y_i, x_{i1}, x_{i2}, \dots, x_{ik}] \quad i = 1, \dots, n$$

Stacked together, we have a matrix of data:

$$\begin{bmatrix} y_1 & x_{11} & x_{12} & \dots & x_{1k} \\ y_2 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}$$

Estimation: An intuitive approach

Suppose you started with the data? For example, the Living Costs and Food Survey: a household level dataset with information on consumption and income.

$$[cons_i, inc_i] \quad i = 1, \dots, n$$

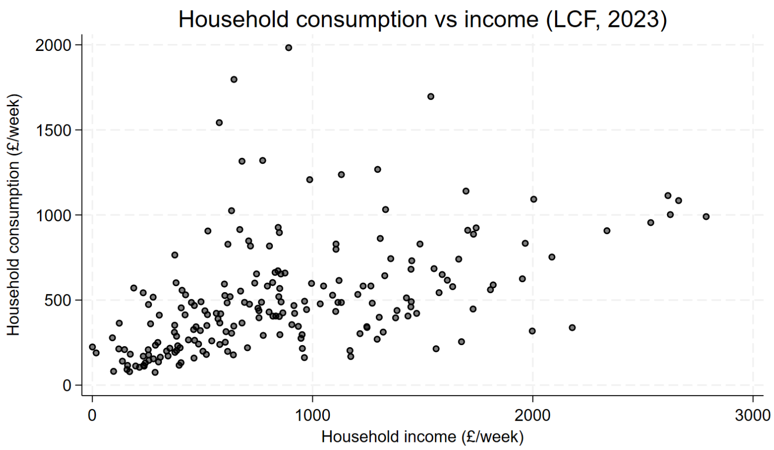
And suppose we had an economic model that suggested a linear relationship between consumption and income (i.e. constant propensity to consume). We might then posit a simple linear regression model:

$$cons_i = \beta_0 + \beta_1 inc_i + u_i$$

Stata

```
* Open data
use "data/lcf_2023.dta", clear

* Plot
scatter hhcon hhinc, mc(black) mfc(%50) ytitle("Household consumption (£/week)") xtitle("Hous
```

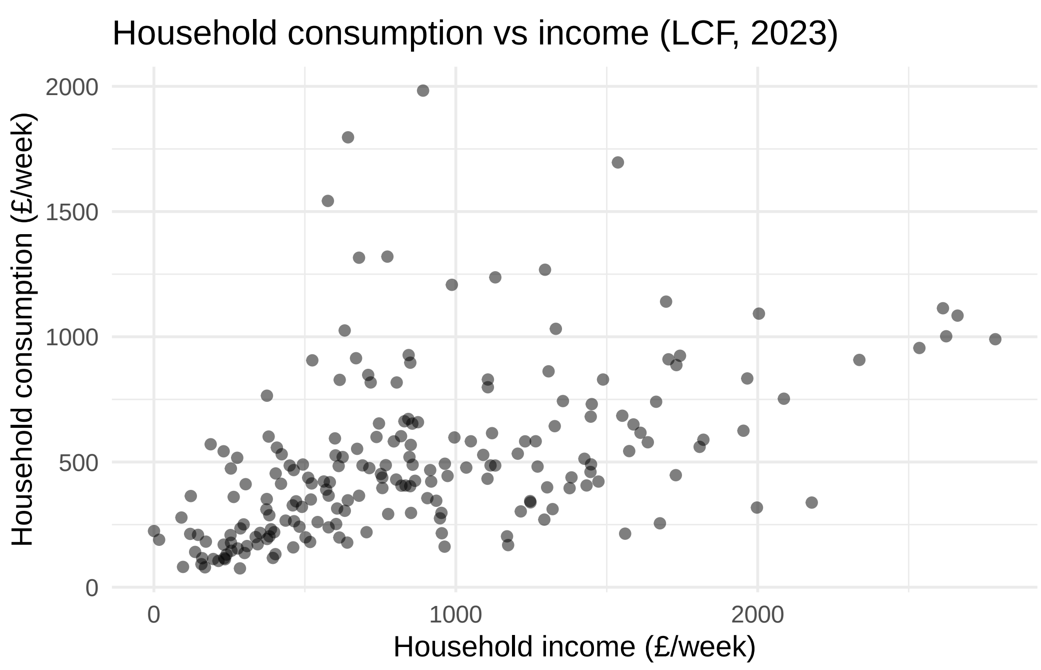


R

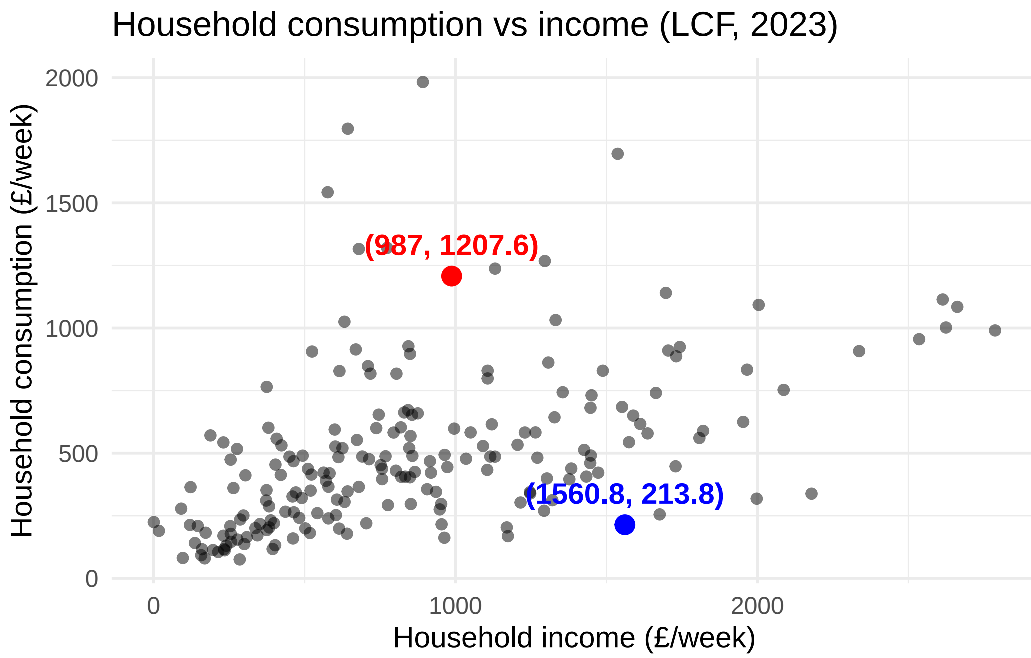
```r
# Packages
library(haven)                    # To load Stata .dta file
library(ggplot2)

# Load the Stata file
lcf_data <- read_dta("data/lcf_2023.dta")

# Plot
ggplot(lcf_data, aes(x = hhinc, y = hhcon)) +
  geom_point(alpha = 0.5) +
  labs(x = "Household income (£/week)",
       y = "Household consumption (£/week)",
       title = "Household consumption vs income (LCF, 2023)") +
  theme_minimal()
```

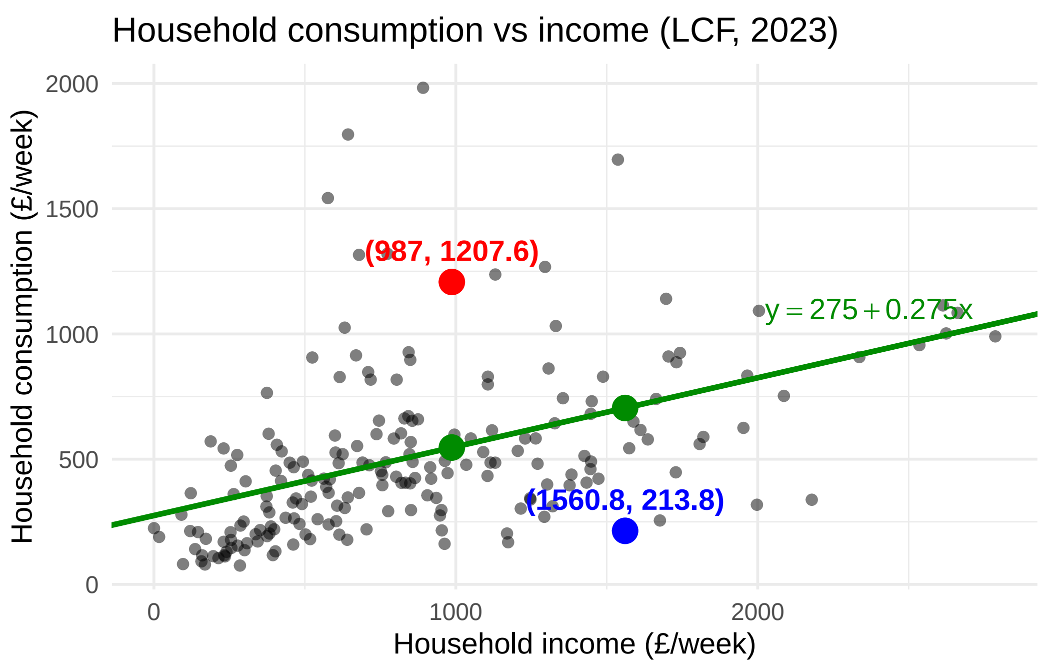


Consider the following two observations:

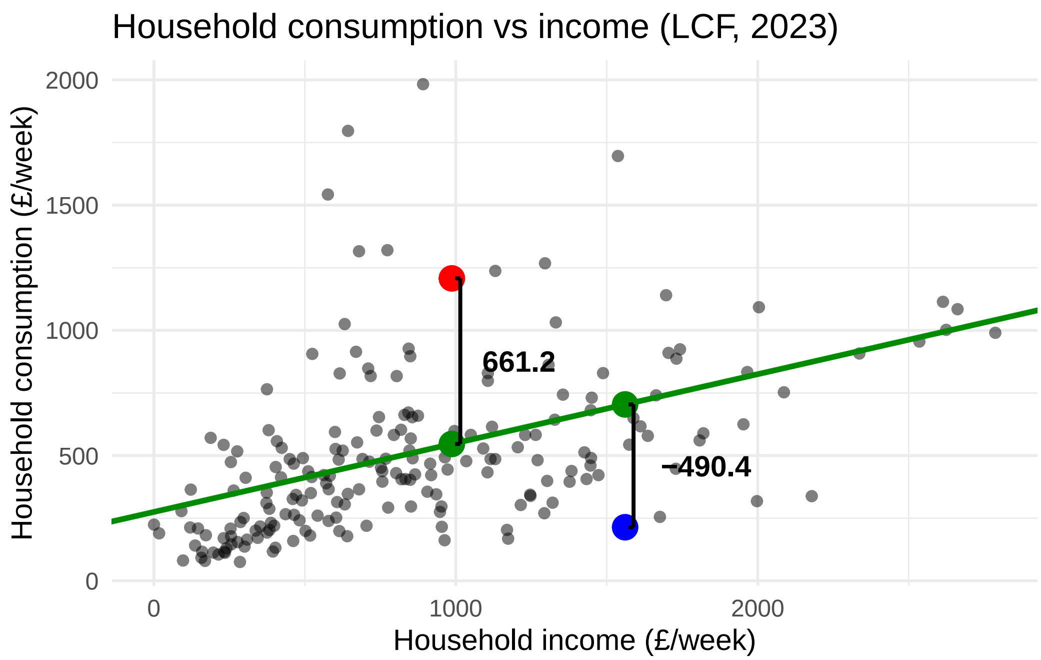


Guess 1

Suppose we plot a line through the points:

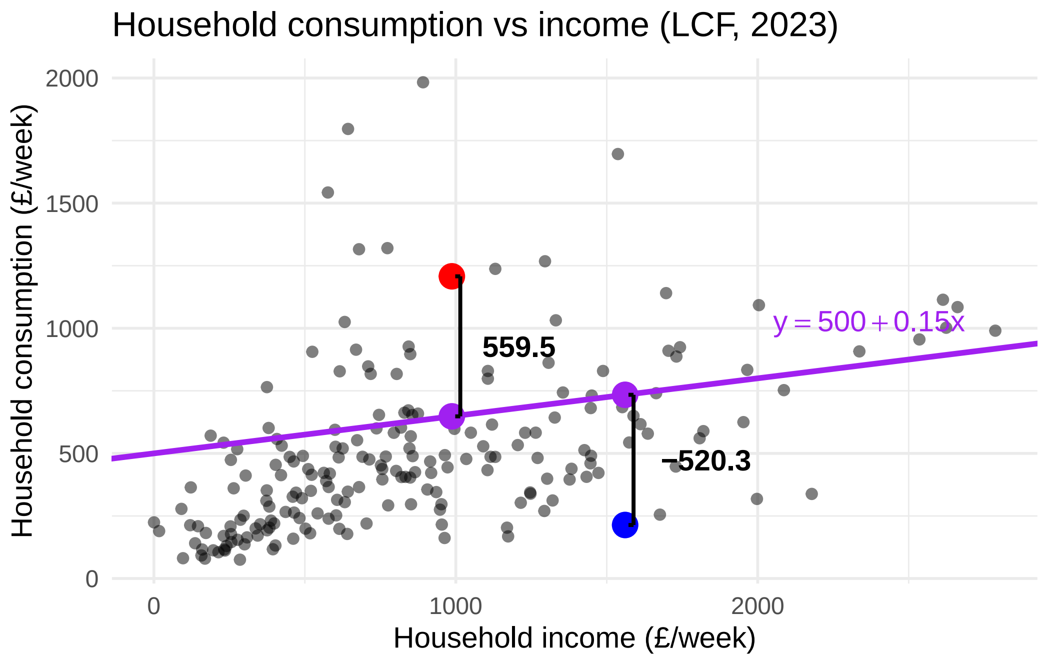


The vertical distance between points in the data and the line tells us how the difference between the “prediction” of the line (for a given inc_i) and the actual data.



Guess 2

What if we chose a different plot?



Which guess was better? Why?

Structure

Let's formalize the above process:

- Goal: we want to pick the line that best fits the data.
- Problem: what do we mean by "fit"?
  - For a given value of inc_i in the data, the "fitted" value of $cons_i$ is

$$\widehat{cons}_i = b_0 + b_1 inc_i$$

  where $[b_0, b_1]$ are the chosen parameters.

  - For example,
    * $\widehat{cons}_i = 275 + 0.275 inc_i$
    * $\widehat{cons}_i = 500 + 0.15 inc_i$

Fit

We need a measure of distance:

Candidate 1: difference

$$cons_i - \widehat{cons}_i$$

- **PROBLEM**: could be negative (i.e. not a measure of distance)

Candidate 2: absolute value

$$|cons_i - \widehat{cons}_i|$$

- **PRO**: intuitive
- **CON**: difficult to work with (i.e. not globally differentiable)

Candidate 3: squared deviation

$$(cons_i - \widehat{cons}_i)^2$$

- always positive
- amplifies bigger deviations, which is more *efficient* (more on that soon!)

Aggregation

What we care about is the difference between the prediction and data across all observations:

$$\sum_{i=1}^{n}(cons_i - \widehat{cons}_i)^2$$

And since $\widehat{cons}_i = b_0 + b_1 inc_i$

$$\sum_{i=1}^{n}(cons_i - b_0 - b_1 inc_i)^2$$

Minization

The value of these squared deviations (from the prediction) depend on $[b_0, b_1]$. It seems intuitive then to pick the values that minimize this value:

$$\min_{b_0,b_1} \quad \sum_{i=1}^{n}(cons_i - b_0 - b_1 inc_i)^2$$

Ordinary Least Squares

Notation

- Data: $[y_i, x_{i1}, x_{i2}, \ldots, x_{ik}] \quad i = 1, \ldots, n$
-

Problem

Minimization problem:

$$\hat{\beta} = \arg\min_{b} \quad \sum_{i=1}^{n}(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik})^2$$

where $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k]$ and $b = [b_0, b_1, \ldots, b_k]$.

- hence the name "Least Squares"

First order conditions

Differentiate w.r.t. b_j's and then set to 0. For $\hat{\beta}_0$ (the constant):

$$\begin{aligned} & 0 = 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \quad & 0 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \quad & \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \cdots - \hat{\beta}_k \bar{x}_k \end{aligned}$$

Note

The inclusion of a constant in the model ensures that the residual is always mean zero:

$$0 = \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = \sum_{i=1}^{n}\hat{u}_i$$

For $\hat{\beta}_j \quad j = 1, \dots, k$

$$\begin{aligned}
0 &= 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})x_{ij} \\
\Rightarrow \quad 0 &= \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})x_{ij}
\end{aligned}$$

There are k of these first-order conditions (plus the 1 for the constant)

$\Rightarrow k+1$ equations and $k+1$ unknowns.

Univariate case

In the case of a single regressor, the solution is more easily solved.

- For $\hat{\beta}_0$ (the constant):

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

- For $\hat{\beta}_1$:

$$\begin{aligned}
0 &= 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i \\
\Rightarrow \quad 0 &= \sum_{i=1}^{n}\left(y_i - (\bar{y} - \hat{\beta}_1\bar{x}) - \hat{\beta}_1 x_i\right)x_i \qquad \text{substitute in} \quad \hat{\beta}_0 \\
\Rightarrow \quad 0 &= \sum_{i=1}^{n}\left((y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})\right)x_i
\end{aligned}$$

The next step explooits the fact that:

$$\sum_{i=1}^{n} \hat{u}_i = 0 \Rightarrow \bar{x} \sum_{i=1}^{n} \hat{u}_i = 0 \Rightarrow \sum_{i=1}^{n} \hat{u}_i \bar{x} = 0$$

Therefore we can minus 0 from the right-hand side without changing anything.

$$\begin{aligned}
0 &= \sum_{i=1}^{n} \big(\underbrace{(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})}_{\hat{u}_i} \big) x_i - \sum_{i=1}^{n} \hat{u}_i \bar{x} \\
\Rightarrow \quad 0 &= \sum_{i=1}^{n} \big((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})\big)(x_i - \bar{x}) \\
\Rightarrow \quad \hat{\beta}_1 &= \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)}
\end{aligned}$$

Sample Analogue

Recall from Lecture 1, under the MLR assumptions (notably, $E[x_i u_i] = 0$)

$$\begin{aligned}
\beta_1 =& \frac{Cov(y_i, x_i)}{Var(x_i)} \\
\beta_0 =& E[y_i] - \beta_1 E[x_i]
\end{aligned}$$

The OLS estimators for a univariate model are the **sample analogue**

$$\begin{aligned}
\hat{\beta}_1 =& \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)} \\
\hat{\beta}_0 =& \bar{y} - \hat{\beta}_1 \bar{x}
\end{aligned}$$

OLS replaces expectations with sample averages.

Decomposition

Given the OLS estimates, we can decompose the outcome into two parts: fitted values ($\hat{y}_i$) and residual ($\hat{u}_i$)

$$y_i = \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\hat{y}_i} + \hat{u}_i$$

where:

- $\sum_{i=1}^{n} \hat{u}_i = 0$: residuals are mean zero
- $\sum_{i=1}^{n} x_i \hat{u}_i = 0$: residuals and regressors are uncorrelated
- $\sum_{i=1}^{n} \hat{y}_i \hat{u}_i = 0$: residuals and fitted values are uncorrelated
- $\sum_{i=1}^{n} \hat{y}_i = \sum_{i=1}^{n} y_i$: fitted values have the same mean as outcome
- $\sum_{i=1}^{n} y_i \hat{u}_i = \sum_{i=1}^{n} (\hat{y}_i + \hat{u}_i)\hat{u}_i = \sum_{i=1}^{n} \hat{u}_i^2$

Example: LCF (2023)

Estimation

Stata

```
* Estimate univariate linear regression
regress hhcon hhinc
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     57.08
       Model |  4842359.39         1  4842359.39   Prob > F        =    0.0000
    Residual |  16798378.5       198  84840.2954   R-squared       =    0.2238
-------------+----------------------------------   Adj R-squared   =    0.2198
       Total |  21640737.9       199  108747.427   Root MSE        =    291.27

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .2705895   .0358165     7.55   0.000     .1999587    .3412203
       _cons |   268.0611   37.42847     7.16   0.000     194.2515    341.8707
------------------------------------------------------------------------------
```

R

```r
# Estimate univariate linear regression
model1 <- lm(hhcon ~ hhinc, data = lcf_data)

# View results
summary(model1)
```

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-519.70 -177.46  -60.08   95.78 1473.71 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 268.06105   37.42847   7.162 1.53e-11 ***
hhinc         0.27059    0.03582   7.555 1.52e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 291.3 on 198 degrees of freedom
Multiple R-squared:  0.2238,	Adjusted R-squared:  0.2198 
F-statistic: 57.08 on 1 and 198 DF,  p-value: 1.515e-12
```

Decomposition

Stata

```stata
* Generate fitted values and residuals
predict fitted
predict residual, resid
```

```stata
* Compute averages
sum hhcon fitted residual
```

```
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
```

```
       hhcon |        200    504.1663    329.7687      75.16    1983.02
      fitted |        200    504.1663    155.9919   268.1639   1022.213
    residual |        200   -1.28e-07    290.5408  -519.6953   1473.709
```

```
* Compute variance-covariance matrix
corr hhcon fitted residual, cov
```

```
(obs=200)

             |    hhcon   fitted residual
-------------+---------------------------
       hhcon |   108747
      fitted |  24333.5  24333.5
    residual |    84414 -.000076    84414
```

R

```
# Get fitted values and residuals
reg_vals <- data.frame(
  hhcon    = lcf_data$hhcon,
  fitted   = fitted(model1),
  resid    = residuals(model1)
)
```

```
# Compute averages
colMeans(reg_vals)
```

```
       hhcon       fitted        resid
5.041663e+02 5.041663e+02 2.611245e-14
```

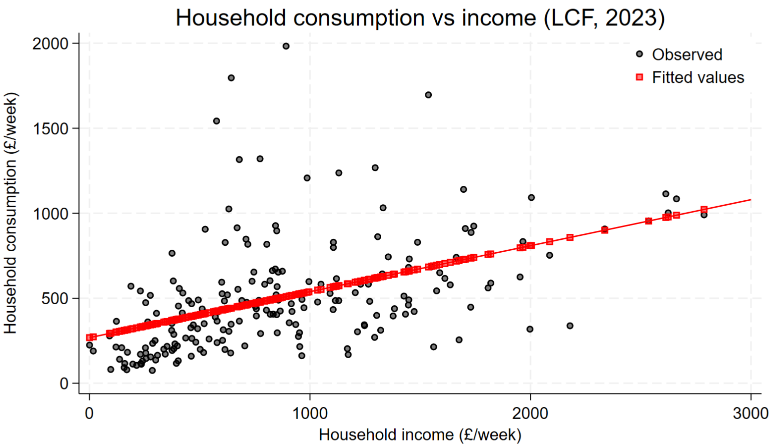
```
# Compute variance-covariance matrix
var(reg_vals)
```

```
           hhcon       fitted        resid
hhcon  108747.43 2.433346e+04 8.441396e+04
fitted  24333.46 2.433346e+04 2.331285e-12
resid   84413.96 2.331285e-12 8.441396e+04
```

Fitted values

Stata

```
* Plot
twoway (scatter hhcon hhinc, mc(black) mfc(%50)) (scatter fitted hhinc, mc(red) mfc(%50) msi:
```

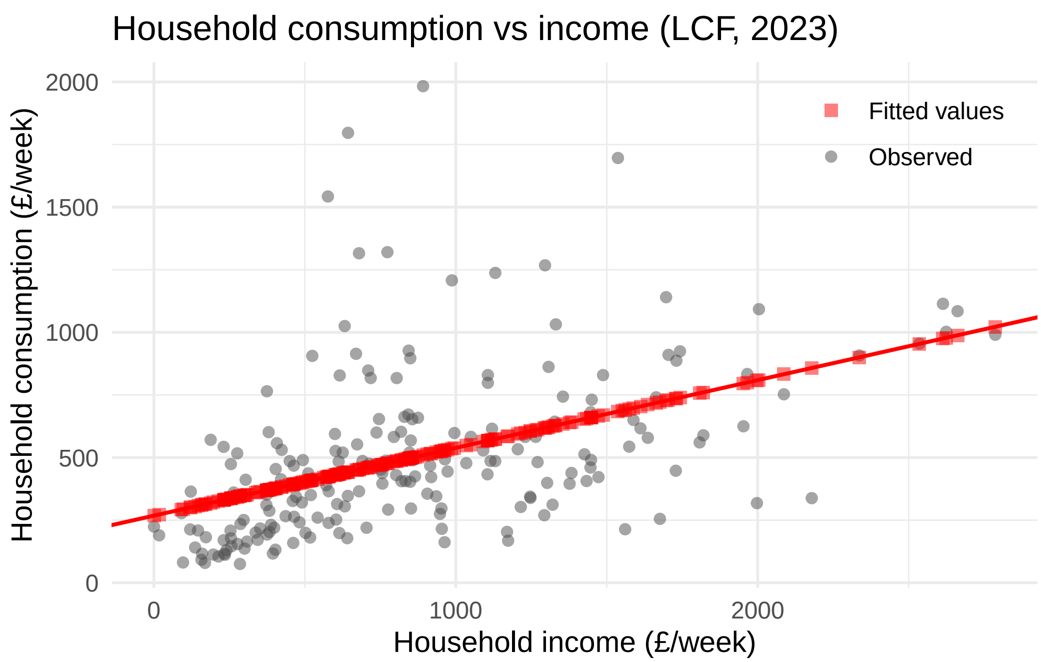


R

```
# Data frame with the x-variable plus fitted values and residuals
plot_vals <- data.frame(
  hhinc  = model.frame(model1)$hhinc,  # exactly the rows lm() used
  hhcon  = model.frame(model1)$hhcon,
  fitted = fitted(model1),
  resid  = residuals(model1)
)

# Plot
ggplot(plot_vals, aes(x = hhinc, y = hhcon)) +
  geom_point(aes(colour = "Observed"), alpha = 0.5) +
  geom_point(aes(y = fitted, colour = "Fitted values"), shape = 15, size = 2, alpha = 0.5) +
  geom_abline(intercept = coef(model1)[1], slope = coef(model1)[2],
              colour = "red", linewidth = 0.7) +
  scale_colour_manual(values = c("Observed" = "grey30",
                                 "Fitted values" = "red")) +
  labs(x = "Household income (£/week)",
```

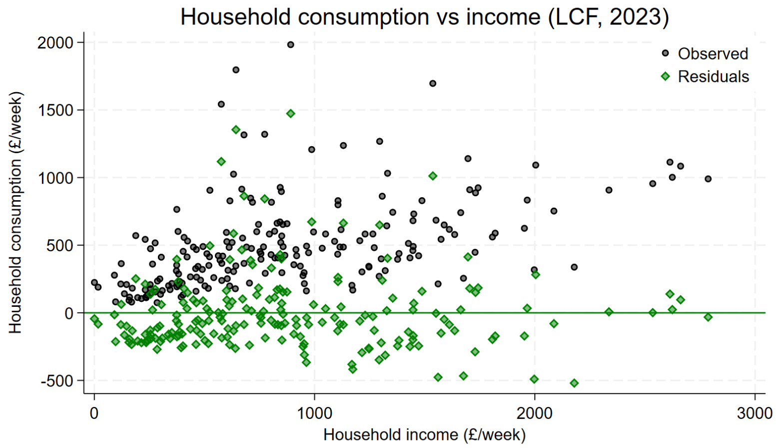
```
      y = "Household consumption (£/week)",
      title = "Household consumption vs income (LCF, 2023)",
      colour = NULL) +
  theme_minimal() +
  theme(legend.position = c(0.98, 0.98),
       legend.justification = c(1, 1))
```



Residual

Stata

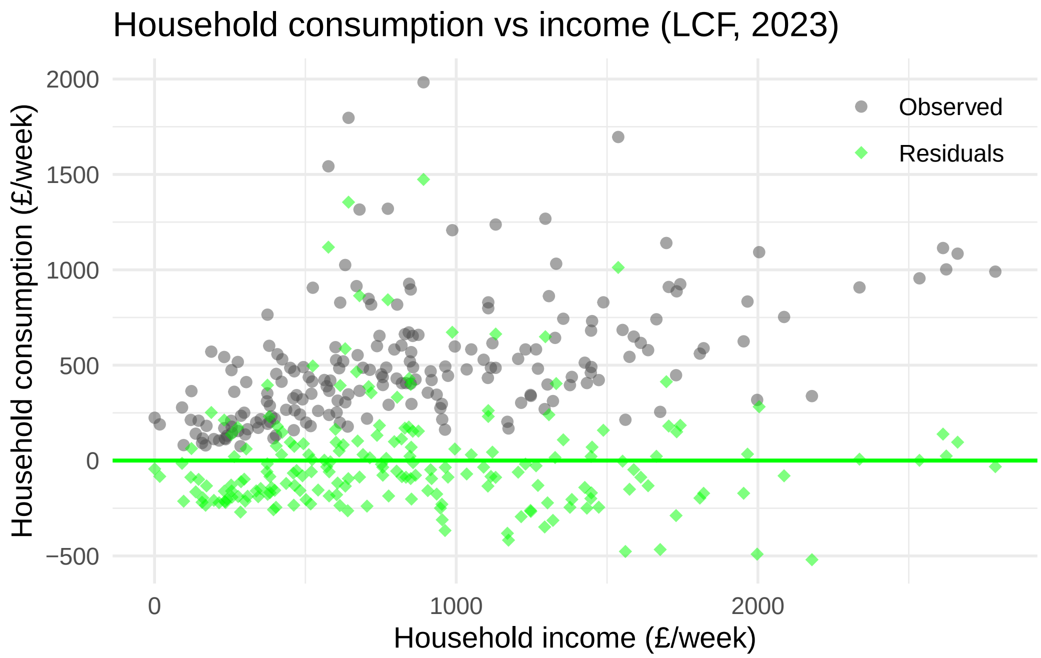
```
* Plot
twoway (scatter hhcon hhinc, mc(black) mfc(%50)) (scatter residual hhinc, mc(green) mfc(%50)
```



R

```
# Data frame with the x-variable plus fitted values and residuals
plot_vals <- data.frame(
  hhinc  = model.frame(model1)$hhinc,  # exactly the rows lm() used
  hhcon  = model.frame(model1)$hhcon,
  fitted = fitted(model1),
  resid  = residuals(model1)
)

# Plot
ggplot(plot_vals, aes(x = hhinc, y = hhcon)) +
  geom_point(aes(colour = "Observed"), alpha = 0.5) +
  geom_point(aes(y = resid, colour = "Residuals"), shape = 18, size = 2, alpha = 0.5) +
  geom_hline(yintercept = 0, colour = "green", linewidth = 0.7) +
  scale_colour_manual(values = c("Observed" = "grey30",
                                 "Residuals" = "green")) +
  labs(x = "Household income (£/week)",
       y = "Household consumption (£/week)",
       title = "Household consumption vs income (LCF, 2023)",
       colour = NULL) +
  theme_minimal() +
  theme(legend.position = c(0.98, 0.98),
        legend.justification = c(1, 1))
```



Goodness-of-Fit

How well did we fit the data?

We have decomposed each observation into two parts:

$$y_i = \underbrace{\hat{y}_i}_{\text{fitted valeus}} + \underbrace{\hat{u}_i}_{\text{residual}}$$

We can apply this decomposition to the *total sum of squares* (SST)

$$\text{SST} \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- The SST is a measure of the total variation in y.

We can show that (Tutorial 1),

$$\mathrm{SST} = \mathrm{SSE} + \mathrm{SSR}$$

where,

- *explained sum of squares* (SSE):

$$\mathrm{SSE} \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- *residual sum of squares* (SSR):

$$\mathrm{SSR} \equiv \sum_{i=1}^{n} \hat{u}_i^2$$

Measure of fit

One key measure of fit then is:

$$\mathbf{R}^2 = \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSR}}{\mathrm{SST}}$$

This measure captures the share of the variance in the outcome variable explained by the regressors.

- If $\mathbf{R}^2 = 1$ then regressors perfectly predict the outcome.

Chasing R^2

Many students make the mistake of thinking that higher $\mathbf{R}^2$ is always better. Here are reasons this need not be the case:

1. In social sciences, the DGP is often very complex and we don't observe many explanatory variables.
    - given the same set of $\mathbf{x}$'s, different outcomes may have very different $\mathbf{R}^2$
2. If we care about the relationship between y and a specific x_j, we do not necessarily care about explaining all of y.

3. We can always add variables and (mechanically) increase $\mathbf{R}^2$; however, this comes at a potential cost:

    a. we *MAY* inadvertantly increase the variance of our estimates (e.g., *irrelevant* variables)

    b. we *MAY* end up adding "bad controls"

Adjusted R^2

An alternative measure fit is the *adjusted* $\mathbf{R}^2$:

$$\overline{\mathbf{R}}^2 = 1 - \frac{\mathrm{SSR}/(n-k-1)}{\mathrm{SST}/n}$$

where,

- n is sample size
- k is the number of regressors $\Rightarrow k+1$ is the number of parameters estimated (including intercept).

As you increase k, you decrease $\overline{\mathbf{R}}^2$

- Creates a disincentive to add irrelevant regressors that do not also reduce SSR.